

PREDICTIVE ANALYTICS FOR STUDENT AT-RISK DETECTION USING RANDOM FOREST

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Year: 2023

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Degree Program: BSIS

Keywords: predictive analytics, student at-risk, Random Forest, academic performance

ABSTRACT

This study builds a Random Forest classifier to predict at-risk students using enrollment, grade, and attendance data, achieving 87.5% accuracy and enabling early intervention programs.

INTRODUCTION

Student dropout and academic failure are costly problems for universities. Early identification of at-risk students enables timely intervention. Traditional methods rely on reactive grade monitoring. This study develops a predictive model using machine learning on historical academic data from CvsU Imus Campus to classify students as At-Risk or Not-At-Risk as early as the 6th week of the semester.

METHODOLOGY

Features used: GWA from previous semester, attendance rate (%), number of incomplete grades, scholarship status, financial aid, and course load. Data covered 1,850 students from AY 2019-2022. Random Forest (100 estimators, max depth 10) was trained on a 75/25 train-test split. SMOTE oversampling addressed class imbalance (25% at-risk). Feature importance was extracted to inform counselor reports. SHAP values were computed for explainability.

RESULTS

Random Forest achieved 87.5% accuracy, precision 84.2%, recall 89.7%, F1 86.9%. The top 3 predictive features were: previous GWA (importance=0.41), attendance rate (0.28), and number of INC grades (0.19). 76 true at-risk students were identified in a live pilot during 2nd semester 2023, and 61 (80%) successfully completed the semester after intervention.

DISCUSSION

The model's 89.7% recall minimizes missed at-risk cases, the higher-priority error in academic intervention contexts. SHAP explanations enabled counselors to understand and trust the predictions. The 80% intervention success rate validates practical utility. Future work should incorporate real-time data integration with the university's SIS for automated early-warning alerts.